

Zero-dimensional Cantor limit sets for hyperbolic finite-adelically distorted orbit counting

Anonymous Authors

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Abstract

The accumulation set of normalized prime-orbit counting is a Cantor set of upper box dimension zero for every genuinely distorted hyperbolic finite-adelically distorted system of positive entropy. Here hyperbolic means that the multiplicative part of the fixed-point sequence has a unique dominant root $\Lambda > 1$, and the normalized count is $N\pi_f(N)/\Lambda^N$, where $\pi_f(N)$ counts primitive orbits of length at most N . The result allows any fixed finite set of distortion primes and the nonzero wild exponential factors in the fixed-point formula. More precisely, if d primes are active, the accumulation set can be covered by at most $C(1 + \log(1/\varepsilon))^{2d}$ intervals of length ε , with constants depending on the fixed system. The proof starts from the known representation of this set as a continuous image of a compact finite-adelic group. An adaptive valuation partition gives the covering bound. A separate Fourier argument for positive sums of radial kernels proves nonconstancy on every admissible cylinder and excludes isolated image values. Both arguments allow nonnegative real exponent data. The theorem addresses the multiprime and wild hyperbolic regimes left open in the cited 2024 public source. Solenoid, additive-group and product examples illustrate the conclusion using classical fixed-point counts.

Keywords: arithmetic dynamics; prime-orbit counting; finite-adelic distortion; Cantor set; box dimension; p -adic Fourier analysis.

1 Introduction and prior work

An orbit-counting function may have the expected exponential order of growth without having a single asymptotic constant. For a map f with finitely many fixed points of every iterate, let $\pi_f(N)$ be the number of primitive periodic orbits of length at most the integer N . Given an exponential growth base $\Lambda > 1$, the observable studied here is the set of accumulation points of

$$\Pi_f(N) = \frac{N\pi_f(N)}{\Lambda^N}, \quad N \in \mathbb{Z}_{\geq 1}. \quad (1.1)$$

Arithmetic factors in the fixed-point sequence can prevent this sequence from converging. Describing all its subsequential limits is then a different question from determining its order of magnitude.

Byszewski, Cornelissen and Houben introduced finite-adelically distorted systems, abbreviated FAD systems, to treat fixed-point formulas with finitely many valuation-dependent factors. Their detector theorem realizes the accumulation set in (1.1) as the image of an explicit continuous function on a compact group [1, Definitions 7.1.1–7.1.2 and Theorem 12.4.3]. Their word *hyperbolic* means that the multiplicative part has a unique dominant root. In that case the archimedean phase disappears, leaving a finite residue coordinate and finitely many p -adic coordinates. For one distortion prime and no wild exponential factor, they prove

Regime	BCH public v2 conclusion	Conclusion here
Hyperbolic, no distortion	Finite accumulation set	Same finite image
Hyperbolic, one prime, no wild term	Finite set united with a Cantor set	Cantor set; bound (1.2)
Hyperbolic, multiprime or wild	Cardinality continuum; topology left open	Cantor set; bound (1.2)
Nonhyperbolic	Contains a nondegenerate interval	Not treated

Table 1: Theorem scope, with the prior-work column referring specifically to [1, Theorems 12.5.1–12.5.2 and Problem 14.1.1]. The distorted rows assume at least one active exponent. The covering estimate is an upper bound, not an asserted sharp scale.

that the image is a union of a finite set and a Cantor set. Problem 14.1.1 in their public version of 19 April 2024 singles out the remaining hyperbolic cases: at least two distortion primes, or a nonzero wild term [1, Theorem 12.5.2, Remark 12.5.3 and Problem 14.1.1].

This article proves that every distorted hyperbolic FAD accumulation set is itself a Cantor set and is metrically thinner than the topological classification alone indicates. If $d \geq 1$ distortion primes are active, then

$$N_\varepsilon(\mathcal{L}_f) \leq C(1 + \log(1/\varepsilon))^{2d} \quad (0 < \varepsilon < \varepsilon_0), \quad (1.2)$$

where N_ε counts intervals of length ε . In particular, $\overline{\dim}_B \mathcal{L}_f = \dim_H \mathcal{L}_f = 0$. The constants are allowed to depend on the fixed system. No uniformity over arbitrarily large prime sets or exponents approaching zero is asserted. Theorem 2.1 proves a standalone image theorem under slightly weaker periodicity hypotheses than the FAD definition. Corollary 6.1 supplies the dynamical conclusion.

The detector viewpoint already has close predecessors. For S -integer dynamics, Everest, Miles, Stevens and Ward control normalized orbit counts through compact-group rotations and obtain a quantitative one-prime injectivity result [3, Theorem 1.1, Lemma 2.4 and Corollary 2.5]. For abelian varieties in positive characteristic, in the unique-dominant-root setting and the branch that is not very inseparable, Byszewski and Cornelissen prove the one-characteristic finite-set/Cantor-set conclusion through a one-variable detector and local injectivity [2, Proposition 9.4 and Theorem 9.5]. These counting reductions and examples are classical inputs to the present problem. The term hyperbolic in this article is always used in the precise FAD sense of [1, Definition 10.3.9], not inferred from the terminology in the title of [3].

Two independent arguments are needed. A continuous image of a profinite space need not remain totally disconnected, and even a zero-dimensional compact set may have isolated points. The covering proof retains only the first L summands of the detector series and refines a valuation tree only along balls containing one of their centers. At depth K this needs $O(LK)$ pieces, instead of all p^K residue classes. Taking both L and K logarithmic in $1/\varepsilon$ proves (1.2).

Perfectness comes from a positive radial-kernel lemma. Each active kernel has strictly negative Fourier coefficients at every nontrivial conductor. Among finitely many exponent types, one has the slowest coefficient decay. High-conductor characters approach one at each fixed ordinary integer, and summability then prevents cancellation among positive translated kernels. Local rescaling gives nonconstancy in every relevant p -adic ball. To apply this argument in several coordinates, the other coordinates are fixed at negative integers. That selection device already occurs in BCH’s proof of their cardinality dichotomy [1, proof of Theorem 12.5.1, pp. 132–133]. The additional conclusion here follows from applying the

Fourier lemma on *every* detector cylinder and using the coprime exponent periods to supply an active center there.

The source comparison is deliberately version-specific. Cornelissen and Park study the Cesàro mean of normalized fixed-point counts and large deviations for orbit decompositions [4, Theorem C and Section 4]. Their reference 6 lists the BCH work as forthcoming at EMS Press in 2026. The final book text was not available in the source audit for this article. Thus the statement that the present proof addresses the open hyperbolic regimes refers to the checked 2024 public version; it is not an unconditional priority claim over subsequent book revisions. This qualification is separate from the mathematical statement proved below.

Section 2 states the image theorem. Sections 3–5 prove its metric and topological parts. Section 6 applies it to genuine FAD systems, including multiprime and wild examples. The final section records the boundaries of the result.

2 A finite-adelic image theorem

For a prime p , write $\mathbb{Z}_p$ for the p -adic integers, v_p for the normalized valuation, and $|z|_p = p^{-v_p(z)}$ for $z \neq 0$. Set $v_p(0) = +\infty$. If $s, t \geq 0$, define

$$H_{p;s,t}(z) = p^{-sv_p(z)-tp^{v_p(z)}} \quad (z \neq 0), \quad (2.1)$$

and prescribe

$$H_{p;s,t}(0) = \begin{cases} 0, & (s, t) \neq (0, 0), \\ 1, & (s, t) = (0, 0). \end{cases} \quad (2.2)$$

An exponent pair is called *active* when $(s, t) \neq (0, 0)$. Every kernel is continuous and takes values in $[0, 1]$. Indeed, the valuation is locally constant away from zero, and an active kernel tends to zero as the valuation tends to infinity. The inactive kernel $H_{p;0,0}$ is the constant function one. Convention (2.2) avoids any interpretation of 0^0 .

Fix the following data throughout Sections 2–5:

- a finite set S of primes and a real number $\Lambda > 1$;
- a positive periodic sequence $(r_n)_{n \in \mathbb{Z}}$;
- for every $p \in S$, nonnegative real periodic sequences $(s_{p,n})_{n \in \mathbb{Z}}$ and $(t_{p,n})_{n \in \mathbb{Z}}$ with a common period m_p satisfying $\gcd(m_p, p) = 1$.

Let $w \geq 1$ be a common period of all the sequences. A prime is active if at least one of its exponent sequences is not identically zero. Inactive primes can be removed from S without changing any of the functions below. We make this removal and write $d = |S|$. The list of exponent pairs at each prime is finite, but a pair is not required to have integer or rational entries.

Give $\mathbb{Z}/w\mathbb{Z}$ the discrete topology and set

$$\mathcal{D} = \overline{\{(n \bmod w, (n)_{p \in S}) : n \in \mathbb{Z}\}} \subseteq (\mathbb{Z}/w\mathbb{Z}) \times \prod_{p \in S} \mathbb{Z}_p. \quad (2.3)$$

This is a nonempty compact metrizable group. On it define

$$\Phi(a, x) = \sum_{\ell=0}^{\infty} \Lambda^{-\ell} r_{a-\ell} \prod_{p \in S} H_{p;s_{p,a-\ell}, t_{p,a-\ell}}(x_p - \ell). \quad (2.4)$$

The indices $a - \ell$ are evaluated by periodicity, so the summand does not depend on a representative of $a \in \mathbb{Z}/w\mathbb{Z}$. Write

$$R = \max_n r_n, \quad B = \frac{R}{1 - \Lambda^{-1}}. \quad (2.5)$$

Each summand of (2.4) is bounded by $R\Lambda^{-\ell}$. The Weierstrass criterion therefore gives uniform convergence and continuity of Φ .

For a nonempty compact $E \subset \mathbb{R}$, let $N_\varepsilon(E)$ denote the least number of closed real intervals of length ε covering E . The upper box dimension is

$$\overline{\dim}_B E = \limsup_{\varepsilon \downarrow 0} \frac{\log N_\varepsilon(E)}{\log(1/\varepsilon)}. \quad (2.6)$$

The term *Cantor set* will mean a nonempty compact, perfect, totally disconnected metric space, here realized as a nowhere dense subset of $\mathbb{R}$. Such a space is homeomorphic to the usual Cantor space.

Theorem 2.1 (Quantitative image theorem). *For the fixed data above, the following statements hold.*

1. *If $d = 0$, then $\Phi(\mathcal{D})$ is the finite set*

$$\left\{ \frac{\sum_{\ell=0}^{w-1} r_{a-\ell} \Lambda^{-\ell}}{1 - \Lambda^{-w}} : a \in \mathbb{Z}/w\mathbb{Z} \right\}. \quad (2.7)$$

2. *If $d > 0$, then $\Phi(\mathcal{D})$ is a Cantor set.*

3. *If $d > 0$, there are $C, \varepsilon_0 > 0$, depending on the fixed data, such that*

$$N_\varepsilon(\Phi(\mathcal{D})) \leq C(1 + \log(1/\varepsilon))^{2d} \quad (0 < \varepsilon < \varepsilon_0). \quad (2.8)$$

Consequently $\overline{\dim}_B \Phi(\mathcal{D}) = \dim_H \Phi(\mathcal{D}) = 0$.

Only positivity and periodicity are assumed for r_n , not the stronger gcd-sequence property used in the dynamical application. The coprimality of the exponent periods with their own prime is used to prove perfectness. The covering estimate does not need that coprimality. These roles will be kept separate in the proof.

3 Adaptive valuation covers

A full partition of $\mathbb{Z}_p$ into residues modulo p^K has exponentially many pieces in K . For the detector series, only distances to a finite list of integer centers matter after truncation. Refining the occupied branches of the residue tree gives a smaller partition.

Lemma 3.1 (Adaptive partition). *Let $E \subset \mathbb{Z}_p$ contain at most L points, and let $K \geq 1$ be an integer. There is a finite clopen partition of $\mathbb{Z}_p$ into at most*

$$1 + (p - 1)LK \quad (3.1)$$

atoms such that, for every $e \in E$, the function $z \mapsto \min\{v_p(z - e), K\}$ is constant on each atom.

Proof. Start with the ball $\mathbb{Z}_p$ of depth zero. For $j = 0, \dots, K-1$, split each current depth- j ball containing a point of E into its p children of depth $j+1$. There are at most L such balls at each depth. Thus at most LK splits occur, and each split increases the leaf count by $p-1$. The terminal leaves form a clopen partition satisfying (3.1).

A terminal ball of depth less than K contains no center. For a center outside a ball, the ultrametric inequality makes its distance from every point of that ball the same: if z, z' lie in the ball and e lies outside, then $v_p(z-e) < v_p(z-z')$, hence $v_p(z'-e) = v_p(z-e)$. On a leaf of depth K , either the center is outside and this argument applies, or the center lies inside and both valuations are at least K . This proves the required constancy of the truncated valuations. $\square$

Lemma 3.2 (Valuation tails). *For each active prime p , there is a constant $c_p > 0$ such that every active exponent pair occurring at p satisfies*

$$\sup_{v_p(z) \geq K} H_{p;s,t}(z) \leq e^{-c_p K} \quad (K \geq 1). \quad (3.2)$$

The same c_p works for all pairs at this fixed prime and all integers K .

Proof. If $s > 0$, then the kernel is at most p^{-sK} on the indicated set. If $s = 0 < t$, then $p^{-tp^K} \leq p^{-tK}$, since $p^K \geq K$. For each active pair choose $s \log p$ in the first case and $t \log p$ in the second. Their minimum is positive because there are finitely many active pairs. It can be used as c_p . At $z = 0$ the kernel vanishes, so the prescribed value obeys the same bound. $\square$

Proof of Theorem 2.1, the covering and dimension assertions. Assume $d > 0$, and let R, B be as in (2.5). For $0 < \varepsilon < 1$ choose

$$L = \max \left\{ 1, \left\lceil \frac{\log(8B/\varepsilon)}{\log \Lambda} \right\rceil \right\}, \quad (3.3)$$

$$K_p = \max \left\{ 1, \left\lceil \frac{\log(8dB/\varepsilon)}{c_p} \right\rceil \right\} \quad (p \in S). \quad (3.4)$$

The tail of (2.4) beginning at $\ell = L$ is bounded uniformly by

$$\sum_{\ell \geq L} R\Lambda^{-\ell} = B\Lambda^{-L} \leq \varepsilon/8. \quad (3.5)$$

Lemma 3.2 and (3.4) also give

$$H_{p;s,t}(z) \leq \frac{\varepsilon}{8dB} \quad \text{if } v_p(z) \geq K_p \text{ and } (s, t) \text{ is active.} \quad (3.6)$$

Apply Lemma 3.1 in coordinate p with centers $\{0, \dots, L-1\}$ and depth K_p . Fix a residue $a \in \mathbb{Z}/w\mathbb{Z}$ and two points $(a, x), (a, y)$ in the same product atom. For $\ell < L$ and $p \in S$, their valuations at center ℓ either agree below K_p or are both at least K_p . In the first case their kernel values are equal. In the second case an inactive kernel is constant one, while two active kernel values both lie in $[0, \varepsilon/(8dB)]$. Therefore

$$\left| H_{p;s_p, a-\ell, t_p, a-\ell}(x_p - \ell) - H_{p;s_p, a-\ell, t_p, a-\ell}(y_p - \ell) \right| \leq \frac{\varepsilon}{8dB}. \quad (3.7)$$

For numbers $u_p, v_p \in [0, 1]$, telescoping the product gives

$$\left| \prod_{p \in S} u_p - \prod_{p \in S} v_p \right| \leq \sum_{p \in S} |u_p - v_p|. \quad (3.8)$$

Indeed, replace one factor at a time; the other factors in each difference have absolute value at most one. By (3.7), the variation of the first L summands of Φ on one product atom is at most

$$\sum_{\ell < L} \Lambda^{-\ell} r_{a-\ell} \frac{\varepsilon}{8B} \leq \varepsilon/8.$$

The two tails contribute at most $\varepsilon/4$ by (3.5). Hence the image of the atom, restricted to $\mathcal{D}$, has diameter at most $3\varepsilon/8 < \varepsilon$. Each nonempty restricted atom is compact, so its image fits in a closed interval of length ε .

The total number of intervals needed is consequently at most

$$w \prod_{p \in S} (1 + (p-1)LK_p). \quad (3.9)$$

All the data are fixed, and both L and every K_p are $O(1 + \log(1/\varepsilon))$ as $\varepsilon \downarrow 0$. Absorbing the finitely many factors in (3.9) into C proves (2.8).

Definition (2.6) now gives upper box dimension zero. For every $\alpha > 0$, the same covers have total α -power of their diameters at most $C(1 + \log(1/\varepsilon))^{2d}\varepsilon^\alpha$, which tends to zero. Thus the Hausdorff dimension is also zero. A nondegenerate interval requires at least a fixed positive multiple of ε^{-1} intervals of length ε to cover it, so $\Phi(\mathcal{D})$ contains no such interval. It is compact and hence closed. Its empty interior makes it nowhere dense. Perfectness, which is not implied by these estimates, is proved below. $\square$

4 Positive radial sums and local nonconstancy

Fix a prime p and give $\mathbb{Z}_p$ its Haar probability measure dz . For $k \geq 1$, use the additive character

$$\chi_k(z) = \exp\left(\frac{2\pi i [z]_{p^k}}{p^k}\right), \quad (4.1)$$

where $[z]_{p^k}$ is the representative in $\{0, \dots, p^k - 1\}$. The conductor of this character is p^k . In particular, for every fixed ordinary integer j , one has $\chi_k(j) \rightarrow 1$ as $k \rightarrow \infty$. The choice of a sequence of characters with this last property will prevent cancellation in a positive sum with integer centers.

Lemma 4.1 (Radial Fourier coefficients). *For an active exponent pair (s, t) , put*

$$h(v) = p^{-sv - tp^v} \quad (v \geq 0), \quad A_{s,t}(k) = \sum_{j \geq k} p^{-j} (h(j-1) - h(j)).$$

Then $A_{s,t}(k) > 0$ and

$$\int_{\mathbb{Z}_p} H_{p;s,t}(z) \overline{\chi_k(z)} dz = -A_{s,t}(k). \quad (4.2)$$

If $t = 0$ and $s > 0$, then

$$A_{s,0}(k) = \frac{(p^s - 1)p^{-(s+1)k}}{1 - p^{-(s+1)}}. \quad (4.3)$$

If $t > 0$ and $s \geq 0$, then

$$A_{s,t}(k) \sim p^{-k-s(k-1)-tp^{k-1}} \quad (k \rightarrow \infty). \quad (4.4)$$

Proof. The sequence $h(v)$ is strictly decreasing to zero. With $\delta_j = h(j-1) - h(j) > 0$, telescoping gives

$$H_{p;s,t}(z) = h(0) - \sum_{j \geq 1} \delta_j \mathbf{1}_{p^j \mathbb{Z}_p}(z). \quad (4.5)$$

This identity holds also at zero, since $\sum_{j \geq 1} \delta_j = h(0)$. The series converges uniformly, its remainder being bounded by the tail of that convergent positive numerical series. The character integral on $p^j \mathbb{Z}_p$ is zero for $j < k$ and is p^{-j} for $j \geq k$. Termwise integration of (4.5) proves (4.2) and strict positivity of $A_{s,t}(k)$.

When $t = 0$, the summand in $A_{s,0}(k)$ equals $(p^s - 1)p^{-(s+1)j}$, giving (4.3) by a geometric sum. When $t > 0$,

$$\frac{h(k)}{h(k-1)} = p^{-s-t(p-1)p^{k-1}} \longrightarrow 0.$$

Thus the first summand of $A_{s,t}(k)$ is asymptotic to $p^{-k}h(k-1)$. The sum of all later summands is bounded by

$$\sum_{j \geq k+1} p^{-j}h(j-1) \leq \frac{p^{-k-1}h(k)}{1-p^{-1}},$$

which is negligible compared with $p^{-k}h(k-1)$. This proves (4.4). $\square$

Lemma 4.2 (Positive translated radial series). *Let T be a finite set of active exponent pairs in $[0, \infty)^2$. Let I be a nonempty finite or countable index set, with ordinary integers j_i , types $\theta_i = (s_i, t_i) \in T$, and positive real numbers b_i such that $\sum_{i \in I} b_i < \infty$. Then the continuous function*

$$F(z) = \sum_{i \in I} b_i H_{p; s_i, t_i}(z - j_i) \quad (4.6)$$

is not constant on $\mathbb{Z}_p$.

Proof. The series is uniformly convergent because its kernels are bounded by one. Among the types actually present, choose $\theta_* = (s_*, t_*)$ by first minimizing t and then minimizing s . Equations (4.3)–(4.4) imply

$$\frac{A_{s,t}(k)}{A_{s_*, t_*}(k)} \longrightarrow \begin{cases} 1, & (s, t) = (s_*, t_*), \\ 0, & (s, t) \neq (s_*, t_*). \end{cases} \quad (4.7)$$

For completeness, if $t_* = 0$, any positive t has faster than geometric decay, and among the types with $t = 0$ the smallest s has the slowest geometric decay. If $t_* > 0$, a larger t contributes a factor $p^{-(t-t_*)p^{k-1}}$ that dominates any geometric factor; at equal t , a larger s contributes $p^{-(s-s_*)(k-1)}$. These observations give all cases in (4.7). The ratios are bounded uniformly over $k \geq 1$ and the finite set of types: each converges, and every denominator is strictly positive.

With $\widehat{F}(\chi) = \int_{\mathbb{Z}_p} F(z) \overline{\chi(z)} dz$, translation and Lemma 4.1 give

$$\frac{\widehat{F}(\chi_k)}{A_{s_*, t_*}(k)} = - \sum_{i \in I} b_i \overline{\chi_k(j_i)} \frac{A_{s_i, t_i}(k)}{A_{s_*, t_*}(k)} \longrightarrow - \sum_{\{i: \theta_i = \theta_*\}} b_i < 0. \quad (4.8)$$

The interchange with the integral follows from uniform absolute convergence. The displayed limit follows from dominated convergence for the counting measure, using the uniform ratio bound, $\sum_i b_i < \infty$, and $\chi_k(j_i) \rightarrow 1$ for each fixed i . The sum on the right is strictly positive before its minus sign, because the selected type is present. Hence $\widehat{F}(\chi_k) \neq 0$ for all sufficiently large k . A constant function has zero integral against every nontrivial character, proving the assertion. $\square$

Lemma 4.3 (Local form). *Let $c_\ell > 0$ for $\ell \geq 0$, with $\sum_{\ell \geq 0} c_\ell < \infty$, and let (s_ℓ, t_ℓ) range over a finite set of nonnegative real exponent pairs, now allowing inactive pairs. Consider*

$$G(z) = \sum_{\ell \geq 0} c_\ell H_{p; s_\ell, t_\ell}(z - \ell).$$

If the ball $a + p^K \mathbb{Z}_p$, with $K \geq 0$ and $0 \leq a < p^K$, contains at least one integer center ℓ whose pair is active, then G is not constant on that ball.

Proof. A center outside the ball has constant valuation distance from every point of the ball, by the ultrametric inequality. Its summand is therefore constant there. An inactive pair contributes a constant on every ball. The sum of these constant terms converges absolutely and is constant.

For an active center in the ball write $\ell = a + p^K j$, where $j \geq 0$, and write $z = a + p^K y$. The kernel rescales as

$$H_{p; s, t}(z - \ell) = p^{-sK} H_{p; s, tp^K}(y - j). \quad (4.9)$$

The identity follows by adding K to the valuation, and the zero convention verifies it at $y = j$ as well. The transformed exponent pairs still form a finite active set. Their coefficients $c_\ell p^{-s_\ell K}$ are positive and summable, since $p^{-s_\ell K} \leq 1$. The active part is nonempty by hypothesis, so Lemma 4.2 proves its nonconstancy on $\mathbb{Z}_p$. Rescaling back proves the result. $\square$

The proofs use nonnegative real exponents directly. In particular, no integrality condition on these exponents or injectivity conclusion from a one-prime detector theorem has been imported.

5 Every detector cylinder has nonconstant image

Lemma 5.1 (The detector group). *Set $\nu_p = v_p(w)$. The group in (2.3) is*

$$\mathcal{D} = \left\{ (a, x) \in (\mathbb{Z}/w\mathbb{Z}) \times \prod_{p \in S} \mathbb{Z}_p : x_p \equiv a \pmod{p^{\nu_p}} \text{ for every } p \in S \right\}. \quad (5.1)$$

Proof. Every diagonal integer point satisfies the congruences, whose solution set is closed. This proves one inclusion. Conversely, take a point satisfying them and prescribe a neighborhood by fixing $a \bmod w$ and finitely many residues $x_p \bmod p^{K_p}$. Increasing the K_p if necessary, assume $K_p \geq \nu_p$. The congruences

$$n \equiv a \pmod{w}, \quad n \equiv x_p \pmod{p^{K_p}} \quad (p \in S)$$

are compatible, since their only overlapping prime powers are those already controlled by (5.1). The Chinese remainder theorem provides an arithmetic progression of integer solutions. Thus every such neighborhood contains a diagonal integer point, proving the other inclusion. $\square$

Proposition 5.2 (Nowhere local constancy). *If $d > 0$, the function Φ is not constant on any nonempty relative open subset of $\mathcal{D}$.*

Proof. By Lemma 5.1, every nonempty relative open subset contains a product cylinder

$$U = \{a\} \times \prod_{p \in S} (b_p + p^{K_p} \mathbb{Z}_p), \quad K_p \geq \nu_p, \quad b_p \equiv a \pmod{p^{\nu_p}}. \quad (5.2)$$

Choose the representatives $0 \leq b_p < p^{K_p}$ and fix any active prime $p_0 \in S$. For every other prime $q \in S \setminus \{p_0\}$, select a negative ordinary integer m_q in its prescribed ball. This is

possible by subtracting a sufficiently large positive multiple of q^{K_q} from b_q . Freeze these coordinates and vary only x_{p_0} in its ball. The resulting slice lies in U by (5.1). On this slice the detector function becomes

$$\sum_{\ell \geq 0} c_\ell H_{p_0; s_{p_0, a-\ell}, t_{p_0, a-\ell}}(x_{p_0} - \ell), \quad (5.3)$$

where

$$c_\ell = \Lambda^{-\ell} r_{a-\ell} \prod_{q \neq p_0} H_{q; s_{q, a-\ell}, t_{q, a-\ell}}(m_q - \ell).$$

Every c_ℓ is strictly positive: $r_{a-\ell} > 0$, and $m_q < 0 \leq \ell$ prevents any argument $m_q - \ell$ from vanishing. Every active kernel is strictly positive away from zero, and an inactive one is identically one. Also $c_\ell \leq R\Lambda^{-\ell}$, so these coefficients are summable.

Because p_0 is active, some residue $n_0 \bmod m_{p_0}$ has a nonzero exponent pair. Coprimality of m_{p_0} with p_0 implies that

$$\ell \equiv b_{p_0} \pmod{p_0^{K_{p_0}}}, \quad a - \ell \equiv n_0 \pmod{m_{p_0}} \quad (5.4)$$

has an ordinary integer solution. Its solution progression contains nonnegative integers. Hence the varying ball contains an active center of (5.3). Lemma 4.3 proves nonconstancy on the slice, and therefore on U . $\square$

The use of negative integers in the other coordinates is already present in [1, proof of Theorem 12.5.1, pp. 132–133]. Here it is combined with Lemma 4.3 and (5.4) to obtain nonconstancy on every cylinder, which is the stronger property needed to exclude isolated values.

Completion of the proof of Theorem 2.1. Suppose first that $d > 0$. The image $E = \Phi(\mathcal{D})$ is nonempty and compact. If y were an isolated point of E , there would be an open real interval J such that $J \cap E = \{y\}$. The nonempty set $\Phi^{-1}(\{y\}) = \Phi^{-1}(J)$ would then be open in $\mathcal{D}$, and Φ would be constant on it. This contradicts Proposition 5.2. Thus E is perfect.

Section 3 proves (2.8) and shows that E is nowhere dense and contains no nondegenerate interval. Every connected subset of $\mathbb{R}$ is an interval, so E is totally disconnected. It is therefore a Cantor set in the stated sense, with the two dimensions already obtained from the covers.

If $d = 0$, the detector group is just $\mathbb{Z}/w\mathbb{Z}$, and the empty product in (2.4) is one. Writing $\ell = j + kw$, with $0 \leq j < w$ and $k \geq 0$, gives

$$\Phi(a) = \sum_{j=0}^{w-1} r_{a-j} \Lambda^{-j} \sum_{k \geq 0} \Lambda^{-kw} = \frac{\sum_{j=0}^{w-1} r_{a-j} \Lambda^{-j}}{1 - \Lambda^{-w}}.$$

This is exactly the finite image in (2.7). $\square$

6 Hyperbolic FAD systems and realized examples

We now state precisely the classical reduction that makes the image theorem an orbit-counting result. No fixed-point realizability assertion is made for arbitrary data in Theorem 2.1.

6.1 The FAD convention and detector reduction

A self-map $f : X \rightarrow X$ is *confined* when $\text{Fix}(f^n)$ is finite for every $n \geq 1$. Write $f_n = \#\text{Fix}(f^n)$ and let $P_f(n)$ be the number of primitive periodic orbits of length exactly n . Thus

$$f_n = \sum_{k|n} k P_f(k), \quad \pi_f(N) = \sum_{n=1}^N P_f(n).$$

All time parameters here are native integer iterates; no time change or alternative normalization is introduced.

A sequence $(a_n)_{n \geq 1}$ is a *gcd sequence* if some $h \geq 1$ satisfies $a_n = a_{\text{gcd}(n,h)}$ for every $n \geq 1$. It is periodic and can be extended to all integer indices by periodicity. Following [1, Definitions 7.1.1–7.1.2], a FAD system is a confined system whose fixed-point counts admit a representation

$$f_n = |\det(A^n - I)| c^n r_n \prod_{p \in S} |n|_p^{s_{p,n}} p^{-t_{p,n}} |n|_p^{-1}, \quad n \geq 1. \quad (6.1)$$

Here A is an integral square matrix, $c > 0$, S is a finite prime set, (r_n) is a positive real gcd sequence, and for each $p \in S$ the sequences $(s_{p,n})$ and $(t_{p,n})$ are nonnegative real gcd sequences whose periods are coprime to p . Their least common period is therefore coprime to p as well. The matrix may have size zero, with determinant factor interpreted as one. These are restrictions on the counts of a genuine self-map, not a claim that every integer sequence of this form is realizable.

Expand the multiplicative part as an exponential polynomial, collecting distinct roots with nonzero coefficients:

$$|\det(A^n - I)| c^n = \sum_j m_j \lambda_j^n, \quad \Lambda = \max_j |\lambda_j|. \quad (6.2)$$

For the systems considered below this expression is nonzero. A root of modulus Λ is dominant. The FAD meaning of *hyperbolic* is that there is exactly one dominant root [1, Definition 10.3.9]. It is then the positive real root Λ ; moreover the normalized leading factor is identically one, not just an unspecified positive constant [1, Lemmas 10.3.2 and 10.3.10]. The entropy in this convention is $\log \Lambda$ [1, Definition 10.3.4]. We assume it is positive, equivalently $\Lambda > 1$, without asserting equality with topological entropy for an arbitrary realization of the count sequence.

Remove inactive primes from the representation and let d be the number remaining. Take w to be a common period of its data and form $\mathcal{D}$ and Φ as in (2.3)–(2.4). The classical detector-image theorem specializes to

$$\mathcal{L}_f := \text{Acc}_{N \rightarrow \infty} \frac{N \pi_f(N)}{\Lambda^N} = \Phi(\mathcal{D}). \quad (6.3)$$

To specify the specialization, the general detector of [1, Definition 12.4.1] also includes the phases $(\lambda_j/\Lambda)^n$ of the dominant roots. Hyperbolicity leaves only $\lambda_1/\Lambda = 1$, so this coordinate is trivial. The factor denoted u in that source is identically one. Its equation (12.4) is consequently exactly (2.4), with the continuous zero values in (2.2). Its Theorem 12.4.3(ii) then gives (6.3). Thus the counting asymptotics and the detector construction in this step are entirely prior input.

Corollary 6.1 (Orbit-limit geometry). *Let (X, f) be a positive-entropy hyperbolic FAD system in the convention above. If $d = 0$, the accumulation set of $N \pi_f(N)/\Lambda^N$ is the finite set in (2.7). If $d > 0$, that accumulation set is a Cantor set and*

$$N_\varepsilon(\mathcal{L}_f) \leq C(1 + \log(1/\varepsilon))^{2d} \quad (0 < \varepsilon < \varepsilon_0) \quad (6.4)$$

for constants depending on the fixed system. In particular $\overline{\dim}_B \mathcal{L}_f = \dim_H \mathcal{L}_f = 0$.

Proof. The data in (6.1) satisfy the hypotheses of Theorem 2.1; the gcd condition is stronger than the periodicity used there. Apply that theorem to the classical equality (6.3). $\square$

6.2 Two distortion primes on a solenoid

Let X be the Pontryagin dual of the discrete additive group $\mathbb{Z}[1/15]$, and let $T : X \rightarrow X$ be dual to multiplication by 2. This is a classical S -integer system, and its fixed-point formula is

$$\# \text{Fix}(T^n) = (2^n - 1) |2^n - 1|_3 |2^n - 1|_5 \quad (n \geq 1) \quad (6.5)$$

by [1, Definition 9.1.2 and Proposition 9.1.4]. In this particular case the formula can also be seen from the finite quotient $\mathbb{Z}[1/15]/(2^n - 1)\mathbb{Z}[1/15]$: localization removes exactly the 3- and 5-parts of $2^n - 1$, and duality identifies its cardinality with the fixed-point count.

The familiar lifting-of-valuations calculation gives

$$v_3(2^n - 1) = \begin{cases} 0, & 2 \nmid n, \\ 1 + v_3(n), & 2 \mid n, \end{cases} \quad v_5(2^n - 1) = \begin{cases} 0, & 4 \nmid n, \\ 1 + v_5(n), & 4 \mid n. \end{cases}$$

For example, in the first active case write $2^n = (2^2)^{n/2}$ and use $v_3(2^2 - 1) = 1$; in the second use $2^n = (2^4)^{n/4}$ and $v_5(2^4 - 1) = 1$. Consequently the FAD data are

$$\begin{aligned} A &= [2], & c &= 1, & \Lambda &= 2, & w &= 4, \\ r_n &= 3^{-1_{2|n}} 5^{-1_{4|n}}, & s_{3,n} &= 1_{2|n}, & s_{5,n} &= 1_{4|n}, & t_{3,n} &= t_{5,n} = 0. \end{aligned} \quad (6.6)$$

The exponent periods 2 and 4 are coprime to 3 and 5 respectively, and r_n is a positive gcd sequence of period 4. The multiplicative part $2^n - 1$ has the unique dominant root 2. Both primes are active. Corollary 6.1 shows that the accumulation set of $N\pi_T(N)/2^N$ is a Cantor set with covering bound $C(1 + \log(1/\varepsilon))^4$. The count (6.5) is classical; the conclusion being applied is the limit-set geometry.

6.3 A genuinely wild additive example

Fix an odd prime p , and consider $U(x) = x^p + x$ on $\overline{\mathbb{F}}_p$, the points of the additive algebraic group. The classical formula is

$$\# \text{Fix}(U^n) = p^{n - p^{v_p(n)}} \quad (n \geq 1). \quad (6.7)$$

It is recorded in [1, Example 7.2.7], which attributes it to Bridy. The following calculation verifies this input directly without assigning novelty to the formula.

Let $F(x) = x^p$ and write $n = p^a m$ with $p \nmid m$. In the commuting composition ring generated by F over $\mathbb{F}_p$,

$$U^n - I = (I + F)^n - I = (I + F^{p^a})^m - I.$$

The first nonzero power of F on the right is F^{p^a} , with coefficient $m \neq 0$ in $\mathbb{F}_p$, and the highest is F^n , with coefficient one. The associated additive polynomial therefore has degree p^n and can be written as

$$(U^n - I)(x) = Q(x^{p^{p^a}}),$$

where Q is an additive polynomial of degree p^{n-p^a} and derivative $m \neq 0$. Thus Q has exactly p^{n-p^a} distinct roots over the algebraic closure. The map $x \mapsto x^{p^{p^a}}$ is bijective there, which proves (6.7), including the case $n = p^a$.

Take the size-zero matrix A , with

$$c = \Lambda = p, \quad r_n = 1, \quad S = \{p\}, \quad s_{p,n} = 0, \quad t_{p,n} = 1, \quad w = 1.$$

The multiplicative part is p^n , and the wild exponent is genuinely nonzero. Corollary 6.1 yields a Cantor accumulation set for $N\pi_U(N)/p^N$ and the bound $C(1 + \log(1/\varepsilon))^2$. The cellular automaton $(x_i)_{i \in \mathbb{Z}} \mapsto (x_{i+1} + x_i)_{i \in \mathbb{Z}}$ on $\mathbb{F}_p^{\mathbb{Z}}$ has the same classical count [1, Example 7.2.6]. Equality of these counts is sufficient for the same orbit-counting conclusion; it is not a claim of conjugacy between the two systems.

6.4 Multiprime and wild distortion together

Take the preceding additive example at $p = 7$ and the product map $V = T \times U$ on $X \times \overline{\mathbb{F}}_7$. Fixed-point counts multiply under products; FAD closure under this operation is also recorded in [1, Example 7.1.3]. Hence

$$\# \text{Fix}(V^n) = (2^n - 1)|2^n - 1|_3|2^n - 1|_5 7^{n-7^{v_7(n)}}. \quad (6.8)$$

Here $A = [2]$ and $c = 7$, so the multiplicative part is $14^n - 7^n$ and has the unique dominant root $\Lambda = 14$. The 3- and 5-data and r_n are those in (6.6); additionally $s_{7,n} = 0$ and $t_{7,n} = 1$. The common period is 4, and the active prime set is $\{3, 5, 7\}$. This single genuine system lies in both the multiprime and the wild regimes. Its accumulation set for $N\pi_V(N)/14^N$ is a Cantor set with

$$N_\varepsilon(\mathcal{L}_V) \leq C(1 + \log(1/\varepsilon))^6$$

and zero upper box dimension. These examples are applications of one image theorem, not separate fixed-point-count discoveries.

7 Scope, limitations, and provenance

The two parts of the proof use different hypotheses. The adaptive cover needs finitely many primes and finitely many exponent types, together with geometric decay of the detector weights. It does not need exponent periods coprime to their own prime. In contrast, the perfectness proof uses positive coefficients to prevent Fourier cancellation, and it uses coprime exponent periods to put an active center in every coordinate ball. Dropping the latter hypothesis can change the topological answer even for one active prime.

Proposition 7.1 (A forbidden-period singleton). *Fix $\Lambda > 1$. Retain the definitions of $\mathcal{D}$ and Φ , but allow $S = \{2\}$, $w = 2$, $r_n = 1$, $t_{2,n} = 0$, and $s_{2,n} = \mathbf{1}_{\{n \text{ odd}\}}$. Then the exponent sequence has forbidden own-prime period 2, and*

$$\Phi(\mathcal{D}) = \{(1 - \Lambda^{-1})^{-1}\},$$

although the prime 2 is active.

Proof. Lemma 5.1, which does not use the exponent periods, gives $\mathcal{D} = \{(a, x) : x \equiv a \pmod{2}\}$. In a term with $a - \ell$ odd, the argument $x - \ell$ is odd. Its valuation is therefore zero and the active kernel $H_{2;1,0}(x - \ell)$ equals one. In a term with $a - \ell$ even, the kernel is inactive and again equals one. Every summand of Φ is thus $\Lambda^{-\ell}$, independently of (a, x) , and summation proves the formula. $\square$

This example belongs to the enlarged analytic data class, not to the FAD class in (6.1). Its image still obeys the covering bound, but it is not perfect. It therefore separates the metric and topological roles of the hypotheses without contradicting Corollary 6.1.

No injectivity, local injectivity, or explicit parametrizing homeomorphism is proved for the detector. Nor does perfectness alone imply that the pushforward of Haar measure has no atoms; no such measure-theoretic assertion is made here. The exponent $2d$ in the logarithmic cover is an upper bound, and its optimality or an exact Hausdorff gauge is left open. Signed or complex weights would require a different noncancellation argument. Infinitely many distortion primes would require a replacement for the finite-product cover.

The theorem also excludes a nontrivial continuous archimedean phase. The nonhyperbolic FAD result that the accumulation set contains a nondegenerate interval belongs to [1, Theorem 12.5.2(i)]; it is outside the present image theorem. For the hyperbolic case, the comparison with open problems remains tied to the cited public v2 of that work. The forthcoming EMS version listed in [4, reference 6] was not obtained for comparison, so an unqualified claim about priority over the final book is not supported by the source audit.

Draft provenance. This is an AI-assisted internal mathematical draft. The fixed-point formulas, detector reduction, earlier topology results, and classical negative-integer slice choice are attributed in the text; the additional arguments are the quantitative cover and positive Fourier proof of whole-cylinder nonconstancy. Internal checks are not human peer review. Human authorship, affiliations, funding, and conflict-of-interest declarations have not been supplied and are not invented here.

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